Dictionaries

Dictionaries

- container object
- mutable
- key-value pairs
  - no indexes, but elements maintain their insertion order
  - key values must be immutable

Dictionary Creation

- Literals
- Dictionary method – dict()
- It is often easier to create an empty dictionary and add entries via user input or from a file

Dictionary Operations

- d[key] - indexing operation retrieves the value associated with the key
- d[key] = value- Adds an entry if the entry does not exist, else modifies the existing entry
- del d[key] - deletes the key from the dictionary
- key in d - Tests for existence in dictionary. If it is, returns True, else False

Dictionary Operations

- d.clear() - Removes all items from the dictionary
- d.get(key, default) - reads the value of the key from the dictionary. If the key does not exist, the default value is returned instead. The default value can be set yourself if necessary

Dictionary Operations

- d1.update(d2) - adds another dictionary to the existing dictionary. d1 keys are overwritten if the same keys exist in d2.
- d1.pop(key,default) - removes and returns the key value from the dictionary. If the key does not exist, return the default

Dictionary Iteration

- Iteration over a dictionary defaults to returning the keys
- We can modify this to get the key, value, or both the key and value

Dictionary Iteration

- d.keys() - returns a view object that yields dictionary keys
- d.values() - returns a view object that yields dictionary values
- d.items() - returns a view object that yields (key, value) tuples

Nested Dictionary

- Dictionaries can store other dictionaries as values